

César Villa

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PROFILE

Mechanical engineering student, pursuing a Master's degree at ETH Zürich focused on fluid mechanics, aeronautics, and aerospace. Demonstrates curiosity, determination, and a persistent drive to learn and overcome challenges, with a strong interest in fluid dynamics, compressible flows, and aerodynamic applications.

EDUCATION

- **Swiss Federal Institute of Technology Zürich (ETH Zürich)** 2025 – 2027
MSc in Mechanical Engineering
Relevant Coursework: Hydrodynamics and Cavitations, Turbulent flows, Aerospace propulsion, Aircraft and Cars Aerodynamics, Orbital Dynamics, Computational multiphase thermal fluid dynamics.
Semester thesis | Durability of Engineering Materials Research Group: CFD modeling with OpenFOAM of reactive multiphase transport in porous media to study corrosion-driven degradation.
 - **École Polytechnique Fédérale de Lausanne (EPFL)** 2022 – 2025
BSc in Mechanical Engineering
Relevant Coursework: Compressible flows, Thermodynamics, Electromagnetism, Mechanics of structures, Mechanical behavior of materials, Industrial production processes, Introduction to AI.
Bachelor thesis | Laboratory of Fluid Mechanics Instabilities (LFMI): Analysis of the fall of a porous object in a viscous fluid using a tracking algorithm and data visualization tools.
 - **École du Flon** 2020 – 2022
EPFL Admission | Contest Exam (Around 15% of admission each year)
 - **Lycée Paul Gauguin, Tahiti** 2017 – 2020
French baccalauréat S
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SKILLS

- **Programming languages:** Python, Matlab, Labview, C, Arduino
 - **Modeling and Simulation:** CATIA, Fusion 360, Shapr3D, PIVLab, Abaqus, OpenFOAM, Ansys (CFD)
 - **Tools:** MS Office Suite, LaTeX, Git
 - **Languages:** French (native), English (fluent), Spanish (intermediate), German (beginner)
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WORK EXPERIENCE

- **Sailing instructor | Campus Nautical Center** 09/2024 – 09/2025
Trained and supervised sailing students, ensuring safety and improving technical skills.
 - **Mentor | EPFL, Lausanne** 08/2024 – 12/2024
Provided guidance and academic support to first-year students at EPFL.
 - **Private mathematics and physics teacher | Lausanne** 09/2023 – 06/2024
Tutored high-school students in mathematics, developing communication and teaching skills.
 - **Catering service | Martinez Hotel, Cannes** 06/2022 – 08/2022
Worked in catering during the summer season, contributing to high-end event service operations.
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OTHER EXPERIENCE

- **Coaching Committee | EPFL:** Responsible for organizing and managing events aimed at supporting student development and collaboration.
- **Competitive sailing | Participation in 3 world championships and 7 international championships:**
Best position at the world championship : 4th
- **Conservatory training | Violin and music theory:** 10 years of practice and the integration of an orchestra.

For more information, visit my portfolio: cesarhanakea.github.io